RETRIEVAL-AUGMENTED VISION-LANGUAGE AGENT FOR CCTVS

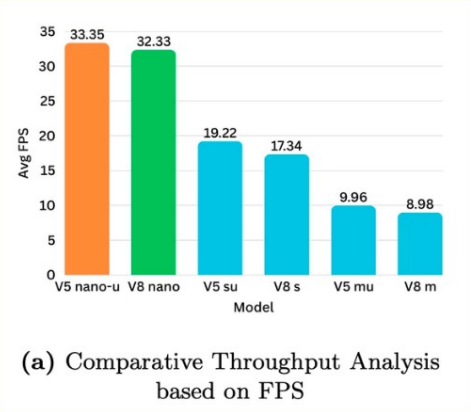
Problem Statement

The aim is to design a system which can convert the CCTV surveillance footage into a searchable, interactive knowledge base. This will increase usability of the collected video archive which can be accessed easily and efficiently for locating events based on the activity.

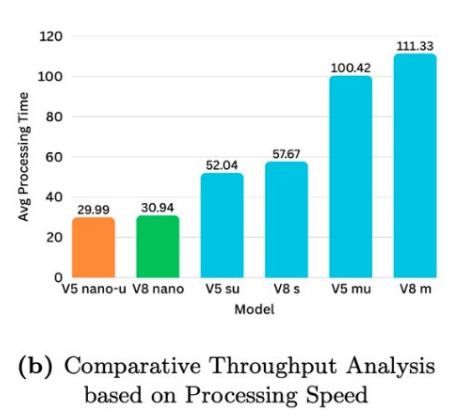
Objectives

- Real-time Detection and Tracking: Initial processing to identify and follow entities across frames.
- Frame Extraction and Processing: Selecting key frames and generating semantic embeddings or captions.
- Natural Language Querying: The final interface allowing users to interact with the stored knowledge base.

Testings



(a) Comparative Throughput Analysis based on FPS



(b) Comparative Throughput Analysis based on Processing Speed

System Architecture

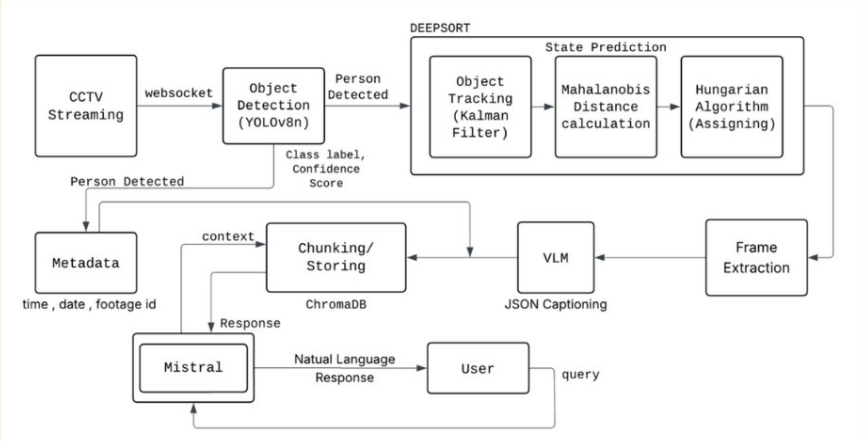


Fig. 4: Illustration of the proposed model.

References

Model	Processing Time	mAP@0.5	Model Size
YOLOv5	32ms	0.82	16.9
YOLOv7	38ms	0.89	72.2
YOLOv8	9.7ms	0.93	8.4

Fig. 1: Model Performance Metrics, 2024 [5]

System Results

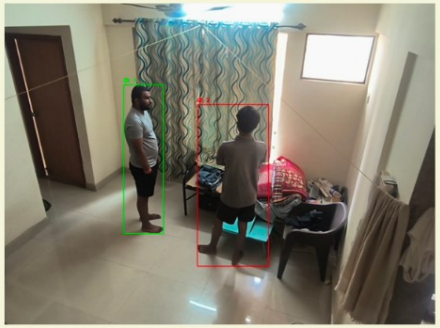


Fig: (Top) Single-object detection and tracking using YOLOv8n+DeepSORT. (Bottom) Multi-object detection on test footage.

Component	Processing Time	Throughput / Metric
YOLOv8n Detection	28-32 ms/frame	30-35 FPS
DeepSORT Tracking	15-82 ms/frame	90% ID retention
VLM Summarization	4-5 s/event	$0.25 < BLEU < 0.32$
ChromaDB Query	42 ms	0.55 to 0.65 recall
RAG Response	1.8 s	100ms latency

Fig: System Component Performance Metrics

Conclusion

The current limitations of this research are its strong reliance on the use of high-end GPU capabilities for VLM inference and its current single-camera architecture, which is not sufficient for wide-area situational awareness.